

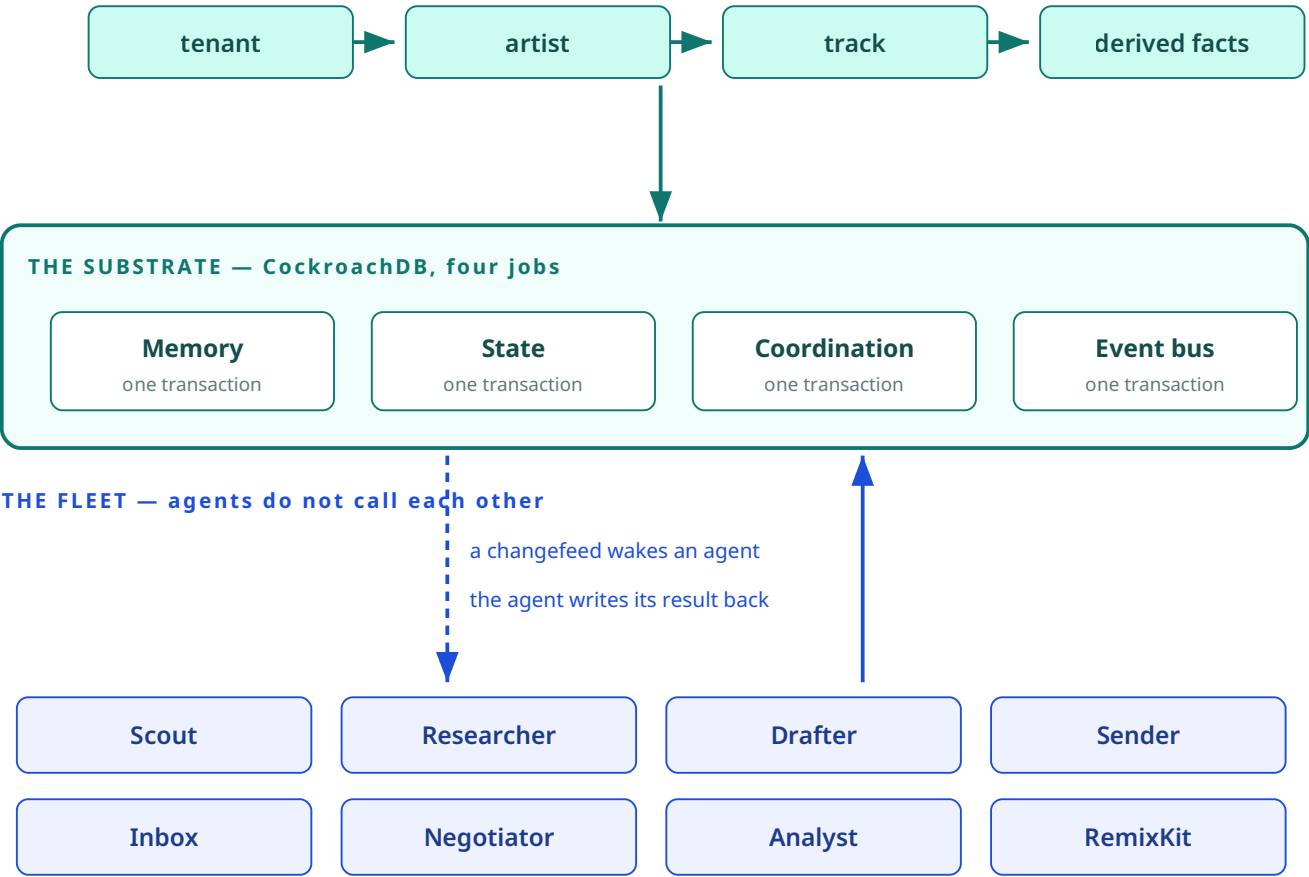
Platform Architecture

One artist's catalogue, many outreach channels, one substrate.

Generated by `infra/platform_architecture.py` · 2026-08-06 · source of record docs/PLATFORM-SPEC.md · submission deadline 2026-08-18 17:00 EDT

Agents do not call each other. They read shared memory, act, write the result back, and that write wakes the next agent. There is no orchestrator, no agent-to-agent RPC and no message broker. A campaign advances because a row changed.

THE SPINE — analysed once



Why one store

JOB	WHAT IT HOLDS	WHERE
Memory	What we know about a track, an artist, a counterparty — and what we have learned	<code>track_character</code> · <code>counterparty_observation</code> · <code>lesson</code>
State	Where every conversation with every counterparty currently stands	<code>campaign</code> · <code>thread</code> · <code>message</code>
Coordination	Which agent owns which work item right now, and for how long	<code>thread.owner_agent</code> · <code>thread.lease_expires_at</code> · <code>outbox</code>
Event bus	Row changes, streamed, as the signal that wakes the next agent	CHANGEFEED FOR <code>thread</code> , <code>outbox</code> , <code>message</code> , <code>counterparty_observation</code>

The conventional stack for these four jobs is Postgres + a vector service + Redis + a queue: four systems, three seams, and every seam is a place two agents can disagree about reality. This is a consolidation and correctness argument — **not** a scale argument. See the verdict on the audit pages.

The data model

Every fact carries how it was obtained. Processes contribute facts; they do not only consume them.

Spine

TABLE	COLUMNS (ABRIDGED)	NOTE
tenant	id, slug, name	
artist	id, tenant_id, slug, name, status	the root — everything compounds here
track	id, tenant_id, artist_id, title, isrc, master_key, status	

Derived facts — analysed once

TABLE	COLUMNS (ABRIDGED)	NOTE
track_measurement	bpm, musical_key, drop_ms, hook_start_ms, energy_curve_json, method	MEASURED · one row, permanent
track_character	mood, era, reference_artists[], embedding VECTOR(1024), model_version	INFERRED · versioned via supersedes_id
track_rights	master_owner, publishing_json, splits_json, clearance_state, asserted_by	ASSERTED · a human is accountable
artist_audience	model_version, position_json, confidence, supersedes_id	INFERRED · inherited by tracks

Counterparty — one shape, five kinds

TABLE	COLUMNS (ABRIDGED)	NOTE
counterparty	kind, platform, handle, bio, profile_embedding VECTOR(1024)	kind: creator radio curator press sync
counterparty_contact	channel, address, role, verified	email dm manager_email
counterparty_observation	dimension, bucket, share, provenance, source, confidence, error_bar_pp	APPEND ONLY · never overwritten

Outreach

TABLE	COLUMNS (ABRIDGED)	NOTE
campaign	artist_id, track_id, channel, goal, state	
thread	campaign_id, counterparty_id, state, next_action_at, owner_agent, lease_expires_at	state machine AND lock, one row
message	direction, channel, subject, body, provider_message_id, idempotency_key	idempotency_key UNIQUE
outbox	kind, payload_json, state, attempts, claimed_by, not_before	the irreversible act, transactionally

Memory and fleet

TABLE	COLUMNS (ABRIDGED)	NOTE
lesson	scope_kind, scope_id, text, evidence_json, embedding VECTOR(1024), supersedes_id	scope: artist counterparty_kind channel global
agent_run	agent_kind, thread_id, input_json, output_json, model, cost_cents, error	makes the fleet restartable, decisions explainable
channel_playbook	channel, state_machine_json, cadence_json, draft_system_prompt, success_metric	a channel is data, not code

Invariants and the loop

Four things the schema makes impossible, and the state machine they protect.

Four invariants the schema enforces

1. Provenance is structural, not conventional

measured / inferred / asserted get three different tables. Mixing them is a schema error, not a discipline failure.

```
track_measurement vs track_character vs track_rights
```

2. Revisions append; they never overwrite

An estimate can never clobber a measurement. The current value is the head of a chain.

```
supersedes_id, on every inferred table
```

3. One open thread per counterparty, across all channels

A creator who is also a curator cannot be worked by two fleets at once. The second fleet's insert fails and does not need to know the first exists.

```
CREATE UNIQUE INDEX ... ON thread (tenant_id, counterparty_id) WHERE state NOT IN ('closed_won', 'closed_lost', 'closed_no_reply')
```

4. A send is recorded in the same transaction that queues it

A crash between 'sent' and 'recorded as sent' would double-email a counterparty — a relationship you burn exactly once.

```
message + outbox written in one txn; idempotency_key UNIQUE
```

Thread state machine — channel-agnostic



The terminal state collapses three: closed_won, closed_lost, closed_no_reply. Only these release the counterparty for a future campaign.

Claiming work — the one primitive every agent uses

```
UPDATE thread
  SET owner_agent = $agent, lease_expires_at = now() + INTERVAL '5 minutes'
  WHERE id IN (
    SELECT id FROM thread
      WHERE tenant_id = $tenant AND state = $state
        AND next_action_at <= now()
        AND (owner_agent IS NULL OR lease_expires_at < now())
      ORDER BY next_action_at LIMIT $batch
    FOR UPDATE SKIP LOCKED)
RETURNING *;
```

A lease, not a lock: an agent that dies mid-task releases its work by expiry with no supervisor involved. This is what makes the fleet restartable — and the demo's closing beat is killing the fleet mid-campaign and watching every thread resume.

The fleet

Eight agents. None of them names another.

Who wakes whom

ROW CHANGE	WAKES	WHICH THEN
thread.state -> shortlisted	Researcher	enriches the counterparty, writes observations
thread.state -> approved	Drafter	retrieves lessons, drafts, sets awaiting_human
thread.state -> queued	Drafter	writes the outbox row — the human has approved
outbox insert	Sender	sends, records provider_message_id
message insert, direction=inbound	Inbox	classifies intent, sets the next state
counterparty_observation insert	Analyst	updates the audience model, may write a lesson

What each agent owns

AGENT	PURPOSE	READS	WRITES
Scout	Find counterparties worth approaching	track_character, counterparty	counterparty, thread(discovered)
Researcher	Enrich one counterparty	counterparty	counterparty_observation
Drafter	Write the outreach	lesson, track_*, counterparty_*	message(outbound), thread
Sender	Perform the irreversible act	outbox	message.sent_at, outbox.state
Inbox	Classify replies, decide next action	message(inbound)	thread.state, lesson
Negotiator	Drive rate and terms to agreement	thread, message, lesson	message(outbound), thread
Analyst	Update the audience model, kill dead hypotheses	counterparty_observation, results	artist_audience, lesson
RemixKit	Generate the assets a brief needs	track_*, artist identity	assets, agent_run

RemixKit is a tool the Drafter calls, not a peer. The topology above is a property of the data, not of any code: no agent imports, addresses or awaits another.

Audit — the verdict

Whether this is the right database, and what has already been decided.

Is CockroachDB the right tool?

Yes for this system, on a narrow argument — and the narrow argument has to be the one presented. At this volume Postgres would serve the workload, and any claim otherwise is false. What is bought is one store instead of four, and correctness defaults that cannot be forgotten. That is a real engineering reason. It is not a scale reason, and presenting it as one would be the dishonest version of this document.

DIMENSION	WINNER	WHY
Scale	NEITHER	Thousands of counterparties, hundreds of sends a day. Neither engine is stretched. Not a differentiator.
Vectors in the same transaction	TIE	pgvector gives this too. It is a reason to avoid Pinecone, not a reason to pick Cockroach.
Serializable by default	COCKROACH	Postgres defaults to Read Committed. Two agents writing lessons about one counterparty silently lose an update. Here the footgun cannot be forgotten — at the cost of retries.
Changefeeds as a first-class feature	COCKROACH	Postgres logical decoding plus Debezium is a project. Here it is one DDL statement.
Scale to zero at \$0 idle	COCKROACH, narrowly	Matches the measured cost model. Neon and Aurora Serverless v2 also scale to zero, so this is a convenience, not a moat.
Ecosystem, ORMs, answers to obscure questions	POSTGRES	PG-wire compatible is not PG-identical. Expect to hit edges.

Decided

DECISION	WHERE	REASONING
The artist is the spine, not the track	SCOPE-RESET §1	Counterparty relationships, the audience model and lessons compound on the artist. A track-rooted schema restarts cold every release.
A track is analysed once; processes query those facts	SCOPE-RESET §2	Five outreach kinds share one counterparty shape, so it is one index and one engine, not six channel tools.
Processes contribute facts, not only consume them	SCOPE-RESET §2a	The failure MEMORY-SPEC diagnosed in RemixKit — read-only memory — must not repeat.
Agents coordinate through the store, not through each other	PLATFORM-SPEC §0	No orchestrator, no agent-to-agent RPC, no broker.
CockroachDB, on consolidation and correctness — not scale	PLATFORM-SPEC §1	At this volume Postgres would serve it. Claiming scale would be the dishonest version.
Two channels by Aug 18: UGC and radio	PLATFORM-SPEC §7	One cannot show the substrate is generic; five multiplies integration surface that is not memory-layer work.

Audit — open, verified, at risk

Everything not yet settled, and everything asserted that has not been checked.

Open

QUESTION	URGENCY	STATE
Counterparty acquisition method	HIGH — gates the Scout	Pillar 10 §4 is a hard 'no scraper, ever'; §5 finds manual sound-page browsing both compliant and higher-signal. Human-in-the-loop bulk acceptance is the presumed middle.
Licence	MEDIUM — required to submit	Repo has none. Apache-2.0 recommended: patent grant, safer commercial default.
Repository topology	LOW — cosmetic until launch	Platform here with RemixKit beneath it, or a new repo with this one vendored.
Tenancy	LOW — policy, not schema	tenant_id is on every table regardless. One label, or a product from commit one.

Verification status

CLAIM	STATUS	SOURCE / ACTION
Vector index available on CockroachDB Basic	VERIFIED	Vendor pricing page, via infra/MEMORY-WORKLOAD.md (2026-07-26)
Basic scales to zero; 50M RU + 10 GiB free per month	VERIFIED	Vendor pricing page; restated on the Basic cluster planning page (2026-08-06)
Changefeeds available on CockroachDB Basic	VERIFIED	Cockroach's own Basic cluster planning page: 'running a changefeed to an external sink, also consumes RUs'. Third-party claims that serverless disables changefeeds date from 2021-22 and are stale. Checked 2026-08-06.
Vector-index filters accelerate on prefix columns with equality/IN only	VERIFIED	Vector index docs: 'Index acceleration with filters is only supported if the filters match prefix columns.' Range comparisons and subqueries fall back to post-filtering. R1 was amended for this — see reference/COCKROACHDB-AI.md.
feature.vector_index.enabled settable on a Basic cluster	UNVERIFIED	GO/NO-GO. Vector indexes require this cluster setting, and cluster settings are generally restricted on serverless tiers. The entire retrieval design depends on it. Check first, on day 1, before anything else is built.
RU cost of a filtered vector scan	UNVERIFIED	Not published anywhere. Stated as headroom, never as spend. Measure on day 1.
RU cost of a continuously running changefeed	UNVERIFIED	Confirmed to consume RUs; the rate is unpublished. Unlike a query this draws continuously, so it is the likelier of the two to erode the free allowance. Measure on day 1 alongside the vector scan.
CockroachDB overage rates past the free tier	UNVERIFIED	Third-party figures only; both vendor pages defer to a quote. Matters only if free breaks.
Improvement in outreach cost from accumulated memory	ASSUMPTION	The one number that would change the conclusion, and the one nothing can produce yet.

Audit — risk register

Ordered by what would cost the most to discover late.

Risks and mitigations

RISK	SEVERITY	MITIGATION
Vector indexes may not be enabable on the Basic tier	HIGH	Requires SET CLUSTER SETTING feature.vector_index.enabled, and serverless tiers restrict cluster settings. If it cannot be set, either it is on by default or the tier is wrong. Day-1 go/no-go, ahead of all other work.
Filtered ANN degrades to post-filtering on non-prefix predicates	MEDIUM	Documented, not speculative. Mitigated by denormalising freshness and open-thread state into counterparty.contact_state, maintained in the same transaction that opens the thread, so every predicate is equality.
Twelve days, greenfield, zero CockroachDB code written	HIGH	Scope expanded to a superset while the clock shrank. Days 5-7 are the core: if everything after slips, the submission still has its headline criterion.
Email deliverability and domain warmup	HIGH	Takes longer than twelve days and scores nothing. The demo sends only to owned and consented addresses; the outbox proves the mechanism without volume.
Free-tier RU erosion by the changefeed	MEDIUM	Continuous draw, unpublished rate. Fallback is polling thread.next_action_at, which is adequate at this volume and costs the architecture its elegance, not its function.
Serializable retries under fleet contention	MEDIUM	Serializable is the reason to choose this database and also a cost: contended transactions retry and the application must handle it. SKIP LOCKED keeps claim contention low, but retry loops are not optional.
Small N on the improvement metric by Aug 18	MEDIUM	Label the curve with its N. Do not draw a flattering one — the house rule that made screen_clips.py abstain applies to our own demo.
Counterparty PII acquires residency obligations	LOW	A creator index is a database of people. If EU counterparties are indexed at volume, REGIONAL BY ROW stops being optional. Not a launch blocker.